

An Introduction to Reinforcement Learning

Caleb Lee, Catherine Cao, Arshia Mago, Ashlee King, Christina Fan^{a,1,*}, Bob Security^{b,2}, Cat Memes^{b,3}, Derek Zoolander

^a*University of Waterloo, Department of Psychology, Street Address, City, Postal Code*

^b*Another University, Department Name, Street Address, City, Postal Code*

Abstract

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Keywords: Q-learning, Reinforcement learning, State, Agent, Action, TD learning

Please make sure that your manuscript follows the guidelines in the Guide for Authors of the relevant journal. It is not necessary to typeset your manuscript in exactly the same way as an article, unless you are submitting to a camera-ready copy (CRC) journal.

For detailed instructions regarding the elsevier article class, see <https://www.elsevier.com/authors/policies-and-guidelines/latex-instructions>

1. Bibliography styles

Here are two sample references: [Feynman and Vernon Jr. \(1963\)](#) [Dirac \(1953\)](#).

*Corresponding author

Email addresses: alice@example.com (Caleb Lee, Catherine Cao, Arshia Mago, Ashlee King, Christina Fan), bob@example.com (Bob Security), cat@example.com (Cat Memes), derek@example.com (Derek Zoolander)

¹This is the first author footnote.

²Another author footnote, this is a very long footnote and it should be a really long footnote. But this footnote is not yet sufficiently long enough to make two lines of footnote text.

³Yet another author footnote.

With this template using `elsevier` class, `natbib` will be used. Three bibliographic style files (*.bst) are provided and their use controlled by `cite-style` option:

- `citestyle: number` (default) will use `elsarticle-num.bst` - can be used for the numbered scheme
- `citestyle: numbername` will use `elsarticle-num-names.bst` - can be used for numbered with new options of `natbib.sty`
- `citestyle: authoryear` will use `elsarticle-harv.bst` — can be used for author year scheme

This `citestyle` will insert the right .bst and set the correct `classoption` for `elsarticle` document class.

Using `natbiboptions` variable in YAML header, you can set more options for `natbib` itself. Example

```
natbiboptions: longnamesfirst,angle,semicolon
```

1.1. Using CSL

If `cite-method` is set to `citeproc` in `elsevier_article()`, then `pandoc` is used for citations instead of `natbib`. In this case, the `cs1` option is used to format the references. By default, this template will provide an appropriate style, but alternative `cs1` files are available from <https://www.zotero.org/styles?q=elsevier>. These can be downloaded and stored locally, or the url can be used as in the example header.

2. Equations

Here is an equation:

$$f_X(x) = \left(\frac{\alpha}{\beta}\right) \left(\frac{x}{\beta}\right)^{\alpha-1} e^{-\left(\frac{x}{\beta}\right)^\alpha}; \alpha, \beta, x > 0.$$

Inline equations work as well: $\sum_{i=2}^{\infty} \{\alpha_i^\beta\}$

3. Figures and tables

Figure 1 is generated using an R chunk.

4. Tables coming from R

Tables can also be generated using R chunks, as shown in Table 1 example.

```
knitr::kable(head(mtcars)[,1:4])
```

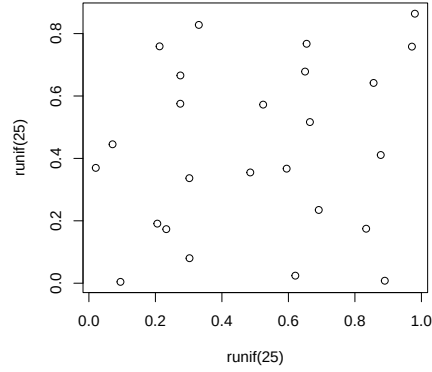


Figure 1: A meaningless scatterplot

Table 1: Caption centered above table

	mpg	cyl	disp	hp
Mazda RX4	21.0	6	160	110
Mazda RX4 Wag	21.0	6	160	110
Datsun 710	22.8	4	108	93
Hornet 4 Drive	21.4	6	258	110
Hornet Sportabout	18.7	8	360	175
Valiant	18.1	6	225	105

5. Caleb Part

- Q learning in computational psychiatry
- Example: D1/D2 receptors

References

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- Feynman, R.P., Vernon Jr., F.L., 1963. The theory of a general quantum system interacting with a linear dissipative system. *Annals of Physics* 24, 118–173. doi:[10.1016/0003-4916\(63\)90068-X](https://doi.org/10.1016/0003-4916(63)90068-X).